Paediatric Antiretroviral Dosages

CHAPTER 4.17	PAEDIATRIC ANTIRETROVIRAL DOSAGES	
Antiretroviral	**Common Dosage**	**Remarks**
Nucleoside reverse transcriptase inhibitor (NRTI)		
Abacavir (ABC)	**Age ≥ 3 months old:** 8 mg/kg/dose (max 300 mg) PO q12H **Adolescent (weight ≥ 25 kg) and adult dose:** 300 mg PO q12H **OR** 600 mg PO q24H	• Not approved for infants < 3 months. • Once daily dosing: 16 mg/kg/dose (max 600 mg) PO q24H.
Emtricitabine (FTC)	**Infant < 3 month:** 3 mg/kg/dose PO q24H **Age ≥ 3 months to 17 years old:** 6 mg/kg/dose (max 240 mg oral solution) PO q24H **Weight > 33 kg (Adult ≥ 18 years):** 200 mg capsule PO q24H	

<table>
<tr><th>CHAPTER 4.17</th><th colspan="2">PAEDIATRIC ANTIRETROVIRAL DOSAGES</th></tr>
<tr><th>Antiretroviral</th><th>Common Dosage</th><th>Remarks</th></tr>
<tr><td>Lamivudine
(3TC)</td><td>Age < 4 weeks old:
2 mg/kg/dose PO q12H

Age ≥ 4 weeks old:
4 mg/kg/dose (max 150 mg) PO q12H</td><td></td></tr>
<tr><td>Tenofovir Disoproxil Fumarate
(TDF)</td><td>Age ≥ 2 years to < 12 years:
8 mg/kg/dose (max 300mg) PO q24H

Tablet (weight-based) Age ≥ 2 years and Weight > 17 kg :
<table><tr><th>Weight</th><th>Dose</th></tr><tr><td>17 to <22 kg</td><td>150 mg PO q24H</td></tr><tr><td>22 to <28 kg</td><td>200 mg PO q24H</td></tr><tr><td>28 to <35 kg</td><td>250 mg PO q24H</td></tr><tr><td>≥ 35 kg</td><td>300 mg PO q24H</td></tr></table></td><td>Screen for HBV status.
Severe acute exacerbation of hepatitis can occur if Tenofovir Disoproxil Fumarate is discontinued.</td></tr>
<tr><td>Zidovudine
(AZT)</td><td>4 kg to < 9 kg :
12 mg/kg/dose PO q12H

9 kg to < 30 kg :
9 mg/kg/dose (max 300mg) PO q12H</td><td>For MTCT prevention in neonates (refer paediatric protocol):<table><tr><th>Week</th><th>Dose</th></tr><tr><td>≥ 30 to < 35 week gestation</td><td>First 2 weeks:
2 mg/kg/dose PO q12H</td></tr></table></td></tr>
</table>

<table>
<tr><th>Antiretroviral</th><th>Common Dosage</th><th>Remarks</th></tr>
<tr><td></td><td>≥ 30 kg (Age ≥ 18 years) :
300 mg PO q12H
OR
180 – 240 mg/m^2 (max: 300 mg) PO q12H</td><td>
<table>
<tr><td></td><td>3 weeks to 6 weeks:
3 mg/kg/dose PO q12H</td></tr>
<tr><td>≥ 35 week gestation</td><td>4 mg/kg/dose PO q12H</td></tr>
</table>
Neonates who cannot tolerate oral dosing, use intravenous zidovudine at 75% of oral dose.
</td></tr>
<tr><td colspan="3">Non-nucleoside reverse transcriptase inhibitor (NNRTI)</td></tr>
<tr><td>Efavirenz
(EFV)</td><td>Children Age ≥ 3 years and Weight ≥ 10 kg :
<table>
<tr><th>Weight</th><th>Dose</th></tr>
<tr><td>7.5 to <15 kg</td><td>200 mg PO q24H</td></tr>
<tr><td>15 to <20 kg</td><td>250 mg PO q24H</td></tr>
<tr><td>20 to <25 kg</td><td>300 mg PO q24H</td></tr>
<tr><td>25 to <32.5 kg</td><td>350 mg PO q24H</td></tr>
<tr><td>32.5 to <40 kg</td><td>400 mg PO q24H</td></tr>
<tr><td>≥ 40 kg</td><td>600 mg PO q24H</td></tr>
</table>
</td><td>Not approved in neonates.
For patients aged 3 months to < 3 years old, optimal dosing is not generally established.</td></tr>
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<tr>
<td>Nevirapine
(NVP)</td>
<td>
Lead in period for 14 days:
<table>
<tr><th>Weight</th><th>Dose</th></tr>
<tr><td>3.5 - 9 kg</td><td>50 mg PO q24H</td></tr>
<tr><td>6 - 9.9 kg</td><td>80 mg PO q24H</td></tr>
<tr><td>10 - 13.9 kg</td><td>100 mg PO q24H</td></tr>
<tr><td>14 - 19.9 kg</td><td>130 mg PO q24H</td></tr>
<tr><td>20 - 24.9 kg</td><td>150 mg PO q24H</td></tr>
<tr><td>≥ 25 kg</td><td>200 mg PO q24H</td></tr>
</table>
Then, if there are no rash or LFT abnormalities after 14 days, see the maintenance dose below:
<table>
<tr><th>Weight</th><th>Dose</th></tr>
<tr><td>3.5 - 9 kg</td><td>50 mg PO q12H</td></tr>
<tr><td>6 - 9.9 kg</td><td>80 mg PO q12H</td></tr>
<tr><td>10 - 13.9 kg</td><td>100 mg PO q12H</td></tr>
<tr><td>14 - 19.9 kg</td><td>130 mg PO q12H</td></tr>
<tr><td>20 - 24.9 kg</td><td>150 mg PO q12H</td></tr>
</table>
</td>
<td>

Lead-in is required in both instances. Start with half (once daily) of the age-related appropriate dose with immediate release for the first 14 days as the lead-in.
If no rash or other adverse effects continue, then on with twice daily.
For MTCT prevention in neonates (refer to paediatric protocol)
Zidovudine in addition to Nevirapine:

Birth weight 1.5 kg to 2 kg:
8 mg/dose PO (NOT per kg)
Birth weight > 2 kg:
12 mg/dose PO (NOT per kg)

6 weeks of Zidovudine plus three doses of Nevirapine given during the first week of life (first dose at birth, 48 hours after first dose, and 96 hours after second dose)

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Antiretroviral	Common Dosage	Remarks
	≥ 25 kg: 200 mg PO q12H **If stable and fully suppressed, the total daily dose may be given as a once-daily (OD) dose.**	
Protease inhibitor (PI)		
Lopinavir/Ritonavir (LPV/r)	**Dose without concomitant use of Efavirenz, Nevirapine, Nelfinavir, and Fosamprenavir.** **Age 14 days to 12 months:** 300 mg / 75 mg LPV/r per m^2 PO q12H Approx 16 mg / 4 mg LPV/r per kg PO q12H (under 12 months) **Age >12 months to 18 years :** 300 mg / 75 mg LPV/r per m^2 PO q12H (max 400 mg /100 mg LPV/r PO q12H) **Weight band dosing for liquid:**	• Different dosing if concomitant use of Efavirenz, Nevirapine or Nelfinavir, Fosamprenavir (Please refer to manufacturer's recommendations).

Antiretroviral	Common Dosage	Remarks
	<table><tr><th>Weight</th><th>Dose</th></tr><tr><td>3.5 - 9 kg</td><td>1 mL PO q12H</td></tr><tr><td>6 - 9.9 kg</td><td>1.5 mL PO q12H</td></tr><tr><td>10 - 13.9 kg</td><td>2 mL PO q12H</td></tr><tr><td>14 - 19.9 kg</td><td>2.5 mL PO q12H</td></tr><tr><td>20 - 24.9 kg</td><td>3 mL PO q12H</td></tr></table>	
Integrase inhibitor (INSTI)		
Dolutegravir (DTG)	**Children Age ≥ 6 years and Weight ≥ 14 kg :** <table><tr><th>Weight</th><th>Dose</th></tr><tr><td>14 - 24 kg</td><td>25 mg PO q24H</td></tr><tr><td>≥ 25 kg</td><td>50 mg PO q24H</td></tr></table>	
Fixed-dose combination (FDC)		
Zidovudine 300 mg /		

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<tr><td>Lamivudine 150 mg
(Combivir®)</td><td><table><tr><th>Weight</th><th>Dose</th></tr><tr><td>14 - 21 kg</td><td>0.5 tab PO q12H</td></tr><tr><td>21 - 30 kg</td><td>0.5 tab PO in the AM + 1 tab PO in the PM</td></tr><tr><td>≥ 30 kg</td><td>1 tab PO q12H</td></tr></table></td><td></td></tr>
<tr><td>Tenofovir Disoporoxil Fumarate 300 mg / Emtricitabine 200 mg
(Tenvir-EM®, Truvada®)</td><td>Adolescent & Weight > 35kg:
1 tab PO q24H</td><td>• Post-exposure prophylaxis: in combination with Dolutegravir.</td></tr>
<tr><td>Tenofovir Disoporoxil Fumarate 300 mg / Emtricitabine 200 mg / Efavirenz 600 mg
(Atripla®, Viraday®)</td><td>Adolescent & Weight > 35kg:
1 tab PO q24H</td><td></td></tr>
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